

DELIVERABLE 0 – PROJECT PROPOSAL

Game Name: Loving Away (working title)

Team: Group G

Team Members: Yiwei Ye

GAME CONCEPT

Loving Away is a multiplayer arena shooter where 2-4 players compete in fast-paced combat. The game features a cute, cozy art style inspired by Animal Party, with physics-based movement that gives characters a satisfying "weight" and momentum feel.

Players control characters from a top-down perspective and use a charging mechanic to launch projectiles at opponents. The longer you hold the shoot button, the farther your projectile flies. When you shoot, your character gets pushed backward slightly. When hit by a projectile, players get knocked back significantly, potentially pushing them toward the arena boundaries where they can be eliminated.

The key feature is the "heavy" character movement - players have momentum and inertia, so you can't instantly change direction. This creates strategic depth as players must predict where opponents will slide to and time their shots carefully.

GAMEPLAY FEATURES

- Charge-and-shoot mechanic: hold the button longer to shoot farther with a higher arc
- Physics-based movement with momentum, sliding, and knockback effects
- Top-down camera view (angled 20-60 degrees)
- Arena boundaries that eliminate players who go too far outside
- Shadow effects showing projectile height for better aiming
- Cute visual style with projectiles that look like hearts, thumbs up, or other friendly emojis

NETWORKING RELATED CHARACTERISTICS

This game is designed specifically to showcase network programming challenges. The real-time combat and physics-based movement mean that any network issues will be immediately obvious to players:

Latency (lag): Delayed hit detection will make projectiles miss moving targets. The charging mechanic requires precise timing, so input delay will be very frustrating. Knockback effects happening late will confuse players about what hit them.

Jitter (unstable connection): Characters will move in jerky, unpredictable ways. Projectile arcs will look wrong. The momentum-based movement will feel broken when updates arrive inconsistently.

Packet Loss: Missing updates will cause characters to suddenly teleport. Projectiles might disappear or appear out of nowhere. Hits won't register properly, making the game unfair.

The game meets the course requirement that network problems must clearly impair gameplay. Unlike turn-based games where lag is less noticeable, this real-time competitive game will feel terrible with poor network conditions.

TECHNICAL APPROACH

Unity Version: 6.2.6f2 (Latest Recommended Support Version)

The game will use a client-server architecture where one player hosts the game and acts as the server. The server is the authority for all game state - it decides where players are, whether projectiles hit, and how knockback is applied.

For the physics-based movement, I'll use a simplified approach instead of Unity's complex physics engine. Characters will move using basic math (velocity, acceleration, drag) which is easier to synchronize over the network and gives consistent results. This approach still creates the "heavy" feel of momentum and sliding, but without the networking headaches of a full physics simulation.

Projectiles follow a simple curved path calculated from the charge amount. The server determines the trajectory and clients just need to know the start point, end point, and how high it goes to draw the same arc.

DEVELOPMENT PLAN

For Mid-Term Demo (November 20):

- Fully working single-player version with movement, shooting, and knockback (bug-free)
- Character movement tuned to feel "heavy" and satisfying

- Basic multiplayer lobby where players can connect
- Player connection system with username selection
- Configuration menu for network settings
- Bonus: Basic synchronization showing that game objects can be sent over the network

For Final Demo (January 9):

- Complete multiplayer game supporting 2-4 players over LAN
- Smooth gameplay with minimal visible lag
- Lobby system with player management
- Polished and balanced gameplay
- Bonus(maybe): Chat functionality in the lobby

Optional Features (if time permits):

- Power-up items that appear during matches (like showing the projectile trajectory path)
- Enhanced visual effects (character leaning into turns, sliding dust particles)
- Network statistics display for debugging

PROJECT SCOPE

This is a fresh project being built from scratch for this course. I am not reusing code or assets from previous projects.

The project scope is designed to be achievable for a solo developer within the semester timeline. The simplified physics approach is specifically chosen to be network-friendly and easier to debug. The small player count (2-4) keeps the networking manageable. The top-down perspective simplifies rendering and camera work. The arena boundaries naturally limit the world size.

By focusing on getting the core gameplay loop working well and stable, then adding networking, and finally polishing with mitigation techniques, the project follows a logical progression that matches the course structure.

WHY THIS GAME IS A GOOD CHOICE

- The simplified physics approach provides the fun "weight" feeling without complex code
- Small player count (2-4) is realistic for debugging and testing alone

- Arena-based combat naturally limits scope
- Network problems will be immediately obvious, meeting course requirements
- The charging mechanic and knockback create interesting gameplay from simple mechanics
- Top-down perspective is easier to implement than 3D character views
- No need for complex animations - the movement and momentum create the fun feel